1. Introduction

1.1 Problem Definition and Motivation

The intersection of computer vision and video game environments has become a fertile ground for artificial intelligence research, particularly in the development of automated game-playing agents. However, before an agent can make strategic decisions, it must first possess a perfect, machine-readable understanding of the game board. The objective of this project is to develop an automated computer vision pipeline capable of extracting complex, dynamic board states from the critically acclaimed deck-building video game, *Inscryption*.

Extracting visual data from Inscryption presents a highly hostile environment for traditional computer vision techniques. Unlike standard digital card games which utilize flat 2D interfaces with perfectly illuminated digital assets, Inscryption takes place on a simulated physical tabletop within a 3D rendered cabin. This design choice introduces several visual complexities:

- Dynamic Lighting: The board is illuminated by different light sources that cast severe gradients and shadows across the cards. A pixel representing "dark ink" on the left side of the board may have a drastically different absolute RGB value than the exact same ink on the right side of the board.
- Stylized Art: The game assets are heavily stylized, low-resolution pixel art textures mapped onto 3D planes. This introduces significant aliasing, artifacting, and 1-pixel structural gaps in the artwork, defeating standard, out-of-the-box Optical Character Recognition (OCR) engines like Tesseract.
- State Changes: Core gameplay mechanics physically alter the cards. The "Submerged" mechanic flips cards into plain cards with no stats and sigil in the center, and “Merge” mechanics create a large gashing sew line that splits a card to halves. Furthermore, while standard sigils are printed in dark ink on light paper, "Sacrifice" and "Totem" mechanics introduce glowing ink on dark or light backgrounds, completely inverting the visual polarity of the targets.

The motivation for this project is to construct a pipeline that can cut through this visual noise to reliably localize cards within the predefined 3D grid, extract numerical stats (Attack and Health), and identify active gameplay Sigils, thereby translating a chaotic 3D screenshot into a structured JSON or dictionary state.

1.2 Background and Related Work

In modern computer vision, the standard approach to object detection and state extraction is the utilization of Deep Learning, specifically CNNs like YOLO or Mask R-CNN. These models are highly robust to lighting changes, partial occlusion, and resolution degradation. However, deep learning models are entirely dependent on massive, meticulously annotated datasets—often requiring thousands of manually labeled examples to achieve high accuracy. Because no such dataset exists for the niche visual environment of Inscryption, a deep learning approach would require a prohibitive amount of manual data entry before pipeline construction could even begin.

Therefore, this project explores a Classical Computer Vision approach utilizing the OpenCV library. Classical computer vision relies on explicit mathematical heuristics, geometric constraints, morphological transformations, and statistical thresholding rather than black-box neural networks. While deep learning is the modern industry standard, a classical approach forces a deeper understanding of the fundamental geometric and statistical properties of the image. This project serves as an exploration into the absolute limits, capabilities, and inherent brittleness of rule-based visual pipelines when deployed in a highly variable, rendered environment.

1.3 Summary of Approach

To overcome the dynamic lighting and overlapping UI elements, the system was designed as a sequential, multi-stage OpenCV pipeline built in Python. Rather than relying on rigid numbers such as static binary thresholds or hardcoded bounding box sizes, the architecture evolved to utilize dynamic relational statistics.

- Card Localization: To detect the presence of a card and its physical state, the system employs a two-stage test. It first utilizes a Canny edge detector to find normal, face-up cards. To solve the "Submerged" mechanic (which causes the card to physically flip face-down, hiding its sharp edges and complex artwork), the system analyzes the standard deviation of pixel intensities in the center of the card slot. This mathematical approach allows the pipeline to successfully differentiate the high-contrast texture of an empty wooden table from the flat, low-contrast, uniform texture of a face-down card back.

- Stat Extraction: To extract Attack and Health numbers across severe lighting gradients, the system calculates local ink mass by finding the 5th and 95th percentiles strictly within the localized stat region, creating a dynamic contrast threshold. To prevent the system from confusing tribal icons (e.g., a wolf paw) for numerical stats, a spatial "Tribal Veto" logic was implemented to evaluate and discard false positives.
- Sigil Localization: The most complex phase involved localizing the central gameplay Sigils. The system utilizes an auto polarity scanner that mathematically detects histogram skew to dynamically adjust to both dark-ink and glowing-ink sigils. To solve the fragmentation of pixel-art, the system deploys extreme morphological dilation to forcefully melt fragmented art, totems, and double-sigils into a single unified anchor point, allowing the pipeline to draw perfectly standardized bounding boxes.

2. Details of the approach

2.1 Card Detection

Before attempting to extract numerical stats or gameplay mechanics, the pipeline must determine if a slot is occupied and assess the physical state of the card. Initially, the system utilized a global Canny edge detector to locate rectangular boundaries. However, Inscryption's submerged mechanic physically flips the card face-down during certain turns, obscuring its ink heavy artwork and stats completely. To reliably detect these cards without false negatives, the pipeline transitioned to statistical texture analysis.

By isolating the central 40% of the ROI , the system converts the region to grayscale and calculates the standard deviation of the pixel intensities. An empty board features highly textured wood grain and painted tribal icons, resulting in a high standard deviation. Conversely, a face-down Submerged card presents a flat, uniform texture. Through iterative testing, the classification threshold was rigorously calibrated.

2.2 Stat Extraction

Extracting Attack and Health values presented severe challenges due to dynamic, localized 3D lighting. A global binary threshold failed entirely, as a shadow on the left side of the board would read as the same absolute RGB value as dark ink on the right side.

To counter localized shadows, the system calculates contrast strictly within the localized Region of Interest (ROI). The system identifies the 95th percentile of pixel intensity and the 5th percentile, the binary threshold is dynamically established at 35% of the distance between these extremes.

A critical false positive emerged during intermediate testing: tribal icons (such as the canine paw) printed in the Attack column possessed an identical "ink mass" to an Attack number. To differentiate between a stat and an icon, a spatial veto system was implemented. The system extracts a second paw_roi directly above the Attack number.

Because the physical distance of 3D light sources altered the mathematical ink mass across different board positions, this veto logic was manually calibrated with strict lower and upper bounds for all eight individual slots.

```python
def evaluate_attack_logic(slot_name, atk_mass, paw_mass):
    if slot_name == "Enemy_0":
        if (atk_mass > 0.22) or (atk_mass < 0.10): return True
        elif (0.14 < atk_mass < 0.21) and (paw_mass < 0.40): return True
        elif (0.20 < atk_mass < 0.22) and (paw_mass > 0.25): return True
        else: return False

    elif slot_name == "Enemy_1":
        if (atk_mass > 0.285) or (atk_mass < 0.205): return True
        elif (0.27 < atk_mass < 0.285) and (paw_mass < 0.27): return True
        elif (0.25 < atk_mass < 0.27) and (paw_mass < 0.30): return True
        else: return False
```

2.3 Sigil Localization

The central gameplay mechanics—Sigils—proved to be the ultimate challenge of the project, exposing the severe limitations of rigid computer vision techniques in dynamic 3D spaces. The architecture for handling sigils underwent a major pivot mid-project due to insurmountable calibration complexities.

Initially, the objective was ambitious: to fully identify and categorize every sigil on the board. The pipeline isolated the sigil Region of Interest (ROI) and deployed an Auto-Polarity scanner to handle the game's changing ink types. Standard sigils use dark ink, while Sacrifice/Totem sigils glow brightly. The create_32x32_signature function mathematically detected this by calculating histogram skew: Skew = $(P_{95} - P_{50}) > (P_{50} - P_5)$; if the bright pixels ($P_{95}$) were further from the median paper value (P_{50}) than the dark pixels (P_5), the system automatically inverted the thresholding logic. Once the ink was isolated, the fragmented pixels were connected using a 5x5 morphological closing operation. To prepare the blob for template matching, the squash_to_32 helper function was written to paste the extracted contour onto a perfect black square canvas before downscaling it to a 32x32 pixel hash, preventing aspect-ratio distortion. This hash was then compared against a custom database.

This approach yielded mixed results. While it successfully categorized some large, standard sigils, it resulted in frequent failures. The fundamental flaw was resolution degradation. Inscryption uses low-resolution pixel art; extracting tiny double-sigils and forcing them into a standardized 32x32 hash destroys their structural integrity, causing the Normalized Cross-Correlation algorithms to fail unpredictably.

Recognizing that refining the template database required an unfeasible amount of calibration within the project's short time window, the objective was pivoted. The system abandoned categorization and focused entirely on reliable localization—simply drawing a unified bounding box to indicate that a gameplay mechanic was present.

The unified locator is based on four main constraints:

- Dilation: To solve the fragmentation of pixel-art, a 21x21 dilation kernel was applied to the isolated ink. This forcefully melted fractured art (like the disconnected infinity loop and dagger of the "Many Lives" sigil) into a single cohesive blob.
- Spatial Gatekeeping: To prevent this massive dilation from bleeding into the Attack and Health numbers, a strict spatial gatekeeper was enforced. The system instantly discarded any detected contour whose geometric center fell outside the middle 40% of the card.

- Duplicate: If a sigil still managed to split into two contours, an 80-pixel radius duplicate blindspot was utilized. The system anchored onto the first detected center point and programmatically deleted any other detections within that 80-pixel radius, guaranteeing only one output.

Finally, the pipeline drew a static 130x130 yellow bounding box centered perfectly on the surviving anchor point, uniformly framing the mechanic.

3. Results

3.1 Experimental Protocol and Dataset

Because this pipeline relies on classical, rule-based computer vision rather than a trained CNN, a traditional "training/validation" split was not utilized. Instead, the experimental protocol was structured into a two-phase "Calibration and Full-Scale Test" methodology. The dataset consisted of 100 high-resolution screenshots captured directly from Inscryption gameplay. Because the board features eight static card slots, this yielded a massive evaluation pool of 800 individual slot ROIs.

- Calibration: Initial testing was conducted on small, curated batches of screenshots specifically chosen for their extreme edge-cases (e.g., heavy 3D shadowing, Gold Nugget ambient light gradients, Submerged cards, glowing totems, and dense tribal icons). These extreme batches were used to manually calibrate the mathematical bounds for the Spatial Gatekeeper, the Tribal Veto, and the slot-specific atk_mass thresholds.
- Accuracy Test: Once the algorithmic logic was locked, the pipeline was deployed against the entire dataset of 100 screenshots (800 ROIs) to evaluate final accuracy, generalization, and robustness against unpredictable lighting shifts.

3.2 Quantitative Evaluation

The pipeline was evaluated sequentially across its three primary objectives: Card Detection, Stat Extraction, and Sigil Processing:

Card Detection - 100% Accuracy

The statistical texture test proved completely flawless. Across all 800 evaluated ROIs, the pipeline successfully identified empty slots, normal face-up cards, and Submerged face-down cards with 100% accuracy. The mathematical approach of evaluating standard deviation completely eliminated the false negatives that plagued early edge-detection attempts. The flat, uniform texture of the face-down card was consistently isolated regardless of localized candle-lighting or ambient game effects.

Stat Extraction - 99.0% Accuracy

The combination of the dynamic local ink mass calculation and the slot-specific Tribal Veto yielded an outstanding 99.0% success rate for extracting numerical stats. The implementation of the paw_roi spatial check effectively eliminated the false positives where tribal paw prints were previously misidentified as Attack numbers. The negligible 1.0% error margin (1 specific card) was strictly constrained to absolute edge cases involving severe, overlapping visual occlusion.

Sigil Classification < 20% Accuracy

The attempt to squash isolated sigils into a 32x32 pixel hash and cross-reference them against a template database using Normalized Cross-Correlation was a quantitative failure. The low-resolution pixel art suffered extreme structural degradation during scaling, rendering the template matching mathematically unviable and producing erratic, misaligned bounding boxes.

Sigil Localization - 84.6% Accuracy

Upon pivoting to a pure localization model—utilizing a 21x21 dilation kernel, auto-polarity skew detection, and an 80 pixel duplicate blindspot—accuracy stabilized drastically. The pipeline successfully deployed the unified bounding box around the correct sigil cluster in 84.6% of the test ROIs.

4. Discussion and Conclusions

4.1 Summary of Main Insights

The primary insight drawn from this extensive exploration is the inherent brittleness of classical, rule-based computer vision when deployed in a dynamic, 3D rendered user interface. Early iterations of this pipeline relied on "magic numbers"—fixed binary thresholds, hardcoded contour aspect ratios, and global edge-density limits. These rigid geometric heuristics resulted in catastrophic algorithmic failures whenever the game engine shifted the localized lighting, cast a 3D shadow, or introduced an ambient color gradient (such as the Gold Nugget illumination).

Stability was only achieved when the pipeline architecture was fundamentally restructured to rely on relational statistics rather than absolute geometry. Replacing global edge-finding with local standard deviation allowed the system to understand the texture of a Submerged card regardless of how dark the game board became; replacing static binary thresholds with dynamic local ink mass arrays allowed the system to successfully separate numerical stats from shadowed paper.

Ultimately, classical computer vision proved highly effective for state detection and spatial localization, provided the constraints were strictly relative to the localized Region of Interest.

4.2 Qualitative Analysis of Negative Results

While the pipeline achieved exceptional accuracy in Card Detection (100%) and Stat Extraction (99.0%), the Sigil processing phases yielded significant negative results. Thoughtful analysis of these failures reveals the mathematical limitations of morphological operations and template matching.

Classification Bottleneck

The sigil classification phase attempt to categorize sigils using Normalized Cross-Correlation (cv2.matchTemplate with TM_CCOEFF_NORMED) failed with 20% accuracy. The root cause of this failure is resolution degradation. Inscryption utilizes low-resolution pixel art. When the pipeline extracted a small, composite double-sigil (e.g., 15x15 pixels) and padded it to a standardized 32x32 pixel hash, the upscaling algorithms introduced severe interpolation artifacts. Because Normalized Cross-Correlation evaluates structural pixel-by-pixel overlap, these scaling artifacts destroy the mathematical integrity of the hash, resulting in failing correlation scores even when comparing visually identical sigils.

Displacement Conflict

In the Sigil Localization attempt, bounding boxes were occasionally rendered off-center or displaced downward on specific cards (e.g., Sparrow, Raven). This is the result of an unavoidable spatial conflict. To prevent the scanner from generating false positives by reading the dark ink of the card's animal artwork, a strict mathematical ceiling was enforced at 70% of the card height. However, "Airborne" wings and specific totems are printed physically higher than this limit. The algorithm acted as a guillotine, clipping the top of the sigil and artificially depressing the calculated geometric center. While a geometric compensation rule was implemented (shifting center upward by 20 pixels if it touched the boundary), this was a rigid fix for a dynamic problem, leading to the observed random displacements

Totem Fragmentation

The pipeline exhibited false negatives when attempting to localize composite "Totem" sigils (e.g., a glowing infinity loop stacked atop a dagger). The architectural solution to fragmented pixel-art was the application of a 21x21 morphological dilation kernel (cv2.dilate), designed to melt disconnected fragments into a single anchor point. However, the 21x21 kernel represents the maximum safe limit. If the kernel size was increased further, the sigil ink would physically expand into the Attack/Health columns, bypassing the Spatial Gatekeeper (< 0.30) and destroying the entire UI parsing logic. Because the physical gap between certain stacked totem fragments occasionally exceeds 21 pixels, the kernel failed to connect them. The Spatial Gatekeeper subsequently filtered out the isolated, undersized fragments, resulting in a complete failure to detect the totem.

4.3 Potential Solutions and Future Work

The failures of the classification database and the morphological dilation limits definitively prove that classical computer vision cannot achieve 100% UI extraction in a rendered environment of this complexity. The mathematical rules required to solve one edge case (e.g., expanding the scan zone for tall sigils) directly cause catastrophic failures in another (e.g., detecting animal artwork).

The definitive solution to these negative results is a transition to deep learning, specifically training a lightweight CNN or YOLO object detection model. A trained CNN does not rely on rigid dilation kernels or absolute bounding constraints; it evaluates holistic contextual features, allowing it to effortlessly classify heavily artifacted, low-resolution pixel art and ignore adjacent animal artwork.

The primary reason a deep learning approach was not utilized in this project was the lack of an annotated training dataset. However, the current OpenCV pipeline directly solves this bottleneck. Because the Locator successfully identifies and tightly frames standard sigils, Submerged cards, and numerical stats with over 84% accuracy, this classical pipeline can now be deployed as an automated data-generation tool. In future work, this script can be run iteratively over hours of captured Inscryption gameplay footage. It will automatically crop, format, and save tens of thousands of dynamic, real-world examples of numbers and sigils under varying 3D lighting conditions. This pipeline will effortlessly generate the massive, perfectly framed dataset required to train a highly accurate CNN, ultimately bridging the gap between classical computer vision heuristics and modern deep learning classification.

4.4 Conclusion

Extracting game-state data from Inscryption presented a formidable computer vision challenge. While the project ultimately pivoted away from its ambitious goal of full sigil classification, the resulting pipeline represents a highly sophisticated, statistically robust localization engine. By successfully neutralizing dynamic 3D lighting, auto-correcting for inverted visual polarity, and applying dynamic texture analysis, this project laid the exact algorithmic groundwork necessary for future machine learning applications in complex digital environments.

References

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